

Subnetting a TCP/IP Network Using the Magic Box

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Interface Technical Training

Anding

0	+	0	=	0
0	+	1	=	0
1	+	0	=	0
1	+	1	=	1

ANDing

IP = 131.226.85.1	10000011	11100010	01010101	00000001
SNM = 255.255.0.0	11111111	11111111	00000000	00000000

Anding Results:

Subnetwork = 131.226.x.x	10000011	11100010	00000000	00000000
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Subnetwork = 131.226.x.x	10000011	11100010	00000000	00000000
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IP = 131.226.50.4	10000011	11100010	00110010	00000100
SNM = 255.255.0.0	11111111	11111111	00000000	00000000

Creating a custom subnet mask

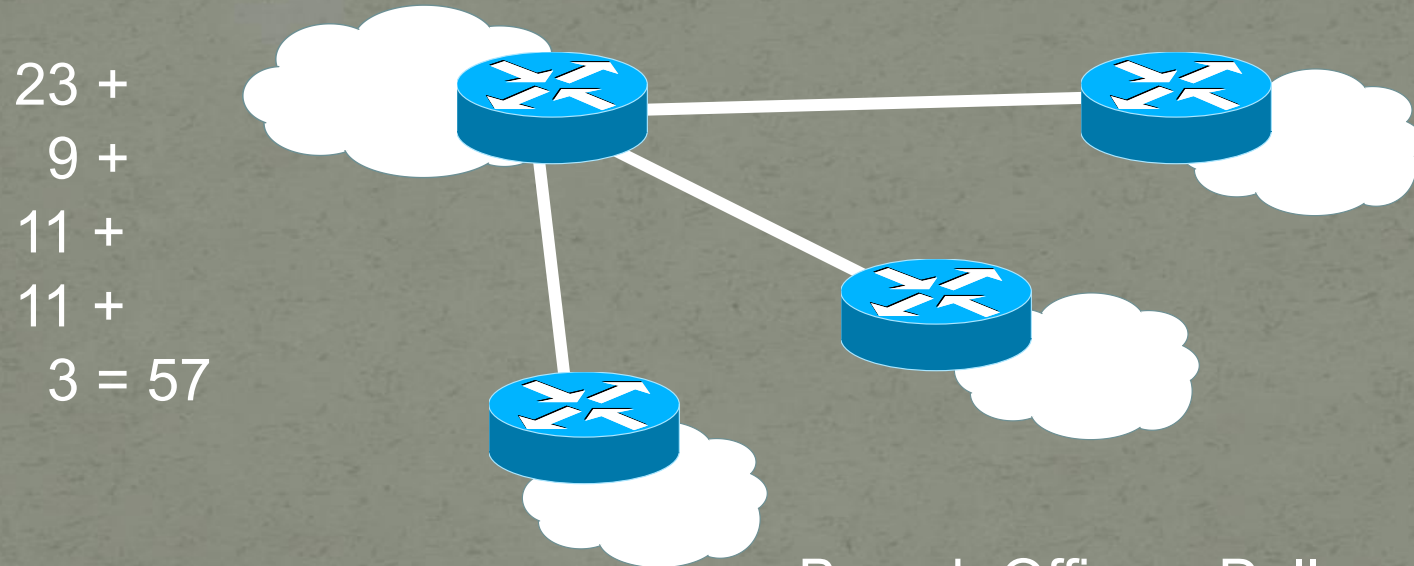
- A custom subnet mask tells TCP/IP to consider some of the bits, which by default would identify a host, as part of the network address
- When a network is divided into subnets, all members of that subnet will have the same network prefix
- Routers divide the networks and provide communication between them

Sample Network Design

Current IP Address 162.16.0.0/16

Central Office – San Jose
23 Ethernet segments

Branch Office – Denver
9 Ethernet segments



Branch Office – Phoenix
11 Ethernet segments

Branch Office – Dallas
11 Ethernet segments

The Magic Box

Top row is your Subnet Mask (high order)

128	192	224	240	248	252	254	255
*	*	*	*	*	*	*	*
128	64	32	16	8	4	2	1
255	127	63	31	15	7	3	1

Bottom row is the number of Networks or Hosts required

Magic Box Process

- 1) Identify Network Address
- 2) Identify Class of Address
- 3) Identify Default Subnet Mask
- 4) Draw the magic line
- 5) Determine the number of required subnets or hosts
- 6) On the bottom line Identify the required number of bits needed to fulfill number 5
- 7) Transpose the number of bits to the top row (This will identify the Subnet mask)
- 8) The number inside the box identifies the network break points

Subnetting Example:

1. Determine IP Address
 - 162.16.0.0
2. Determine Class of IP Address
 - 162.16 is by a Class B
3. Determine default Subnet Mask & convert to Binary
 - 255 . 255 . 0 . 0
11111111.11111111 . 00000000.00000000

Subnetting Example:

4. Draw Magic Line

11111111.11111111.00000000.00000000



5. Sample design indicates accommodation of 57 segments (Including WAN) with no more than 100 hosts per segment (Including router interfaces)

Determine # of bits required

128	192	224	240	248	252	254	255
*	*	*	*	*	*	*	*
128	64	32	16	8	4	2	1
255	127	63	31	15	7	3	1

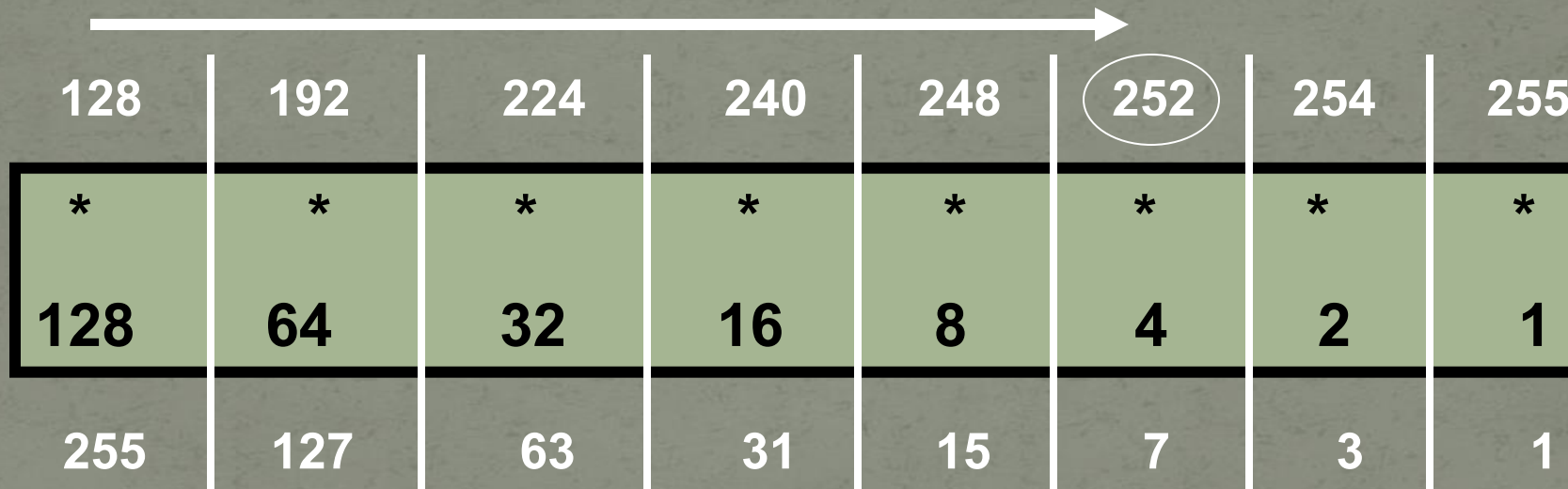
6. Bottom row is the number of networks supported

6 Bits required to support 57 Networks

Determining the new SNM

Top row is your mask (high order)

7. Transpose # of bits to top row



A diagram illustrating the process of transposing the number of bits to the top row of a table. A white arrow points from the left to the right, positioned above the table. The table has 8 columns and 3 rows. The top row contains bit counts: 128, 192, 224, 240, 248, 252 (circled), 254, and 255. The middle row contains asterisks (*). The bottom row contains bit counts: 255, 127, 63, 31, 15, 7, 3, and 1. The columns are separated by vertical white lines. The middle row is highlighted with a green background and a black border.

128	192	224	240	248	252	254	255
*	*	*	*	*	*	*	*
128	64	32	16	8	4	2	1
255	127	63	31	15	7	3	1

– New SNM = 255.255.252.0

8. Finding the Magic Number

128	192	224	240	248	252	254	255
*	*	*	*	*	*	*	*
128	64	32	16	8	4	2	1
255	127	63	31	15	7	3	1

1	1	1	1	1	1	0	0	
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	0	2
0	0	0	0	0	0	1	1	3
0	0	0	0	0	1	0	0	4
0	0	0	0	0	1	0	1	5
0	0	0	0	0	1	1	0	6
0	0	0	0	0	1	1	1	7
0	0	0	0	1	0	0	0	8
0	0	0	0	1	0	0	1	9
0	0	0	0	1	0	1	0	10
0	0	0	0	1	0	1	1	11
0	0	0	0	1	1	0	0	12
0	0	0	0	1	1	0	1	13
0	0	0	0	1	1	1	0	14
0	0	0	0	1	1	1	1	15
0	0	0	1	0	0	0	0	16

Subnetting Example:

- Now use the incremental value to determine the actual network ids, host range and broadcast address:

<i>Network</i>	<i>Host range</i>	<i>Broadcast</i>
162.16.0.0	162.16.0.1 – 162.16.3.254	162.16.3.255 (*zero subnet)
162.16.4.0	162.16.4.1 – 162.16.7.254	162.16.7.255
162.16.8.0	162.16.8.1 – 162.16.11.254	162.16.11.255
162.16.12.0	162.16.12.1- 162.16.15.254	162.16.15.255
162.16.16.0	162.16.16.1- 162.16.19.254	162.16.19.255

and so on, until the last address range

162.16.248.0 – 162.16.251.255
the next number would be the subnet mask

Determine the # of hosts

- Count the number of bits available for host in the new SNM
 - New SNM 255.255.252.0 or 11111111.11111111.11111100.00000000
 - 10 bits available this will be the n^{th} power
- Formula
 - $2^n - 2 = \# \text{ of hosts}$
 - Host cannot be all 0s or all 1s
 - $2^{10} - 2 = \# \text{ of hosts}$
 - $1024 - 2 = 1022$